

DDL Scripts

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DROP SCHEMA IF EXISTS bigbull CASCADE;
CREATE SCHEMA bigbull;
SET search_path TO bigbull;

-- 1. BASE TABLES
CREATE TABLE AppUser (
    UserId VARCHAR(50) PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Email VARCHAR(100) UNIQUE NOT NULL,
    PhoneNumber VARCHAR(20) UNIQUE,
    Address VARCHAR(255),
    PanNumber VARCHAR(20) UNIQUE,
    KycStatus VARCHAR(20) CHECK (KycStatus IN ('Verified', 'Pending',
'Unverified'))
);

CREATE TABLE Company (
    CompanyName VARCHAR(100) PRIMARY KEY,
    Sector VARCHAR(50) NOT NULL
);

-- 2. ACCOUNTS
CREATE TABLE DematAccount (
    DematAccId VARCHAR(50) PRIMARY KEY,
    LinkedDepository VARCHAR(50),
    DateOpened DATE DEFAULT CURRENT_DATE,
    UserId VARCHAR(50) UNIQUE NOT NULL,
    FOREIGN KEY (UserId) REFERENCES AppUser(UserId) ON DELETE CASCADE
);

CREATE TABLE TradingAccount (
    TradingAccId VARCHAR(50) PRIMARY KEY,
    AvlBalance DECIMAL(15,2) DEFAULT 0.00,
    UserId VARCHAR(50) UNIQUE NOT NULL,
    FOREIGN KEY (UserId) REFERENCES AppUser(UserId) ON DELETE CASCADE
);

CREATE TABLE LinkedBankAccount (
    BankAccountNo VARCHAR(30) PRIMARY KEY,
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    IFSC_Code VARCHAR(15) NOT NULL,
    BankName VARCHAR(100),
    IsPrimary BOOLEAN DEFAULT FALSE,
    UserId VARCHAR(50) NOT NULL,
    FOREIGN KEY (UserId) REFERENCES AppUser(UserId) ON DELETE CASCADE
);

-- 3. MARKET
CREATE TABLE Financial_Instruments (
    Ticker_Symbol VARCHAR(20) PRIMARY KEY,
    Instrument_Type VARCHAR(50),
    Exchange VARCHAR(20),
    CompanyName VARCHAR(100) NOT NULL,
    FOREIGN KEY (CompanyName) REFERENCES Company(CompanyName)
);

CREATE TABLE MarketData (
    Ticker_Symbol VARCHAR(20) NOT NULL,
    TimeStamp TIMESTAMP NOT NULL,
    DailyHigh DECIMAL(10,2),
    DailyLow DECIMAL(10,2),
    CurrPrice DECIMAL(10,2),
    Volume BIGINT,
    PRIMARY KEY (Ticker_Symbol, TimeStamp),
    FOREIGN KEY (Ticker_Symbol) REFERENCES Financial_Instruments(Ticker_Symbol)
);

-- 4. PORTFOLIO
CREATE TABLE Portfolio_Holds (
    DematAccId VARCHAR(50) NOT NULL,
    Ticker_Symbol VARCHAR(20) NOT NULL,
    Current_Quantity_Held INT DEFAULT 0,
    PRIMARY KEY (DematAccId, Ticker_Symbol),
    FOREIGN KEY (DematAccId) REFERENCES DematAccount(DematAccId),
    FOREIGN KEY (Ticker_Symbol) REFERENCES Financial_Instruments(Ticker_Symbol)
);

-- 5. TRADING
CREATE TABLE Orders (
    Order_ID VARCHAR(50) PRIMARY KEY,
    Order_Type VARCHAR(20) CHECK (Order_Type IN ('BUY', 'SELL')),
    Price DECIMAL(10,2),
    Status VARCHAR(20) CHECK (Status IN ('Pending', 'Executed', 'Cancelled')),
    OrderQty INT NOT NULL,

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    TimeStamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    TradingAccId VARCHAR(50) NOT NULL,
    Ticker_Symbol VARCHAR(20) NOT NULL,
    FOREIGN KEY (TradingAccId) REFERENCES TradingAccount(TradingAccId),
    FOREIGN KEY (Ticker_Symbol) REFERENCES Financial_Instruments(Ticker_Symbol)
);

CREATE TABLE Trade (
    Trade_ID VARCHAR(50) PRIMARY KEY,
    Execution_Qty INT NOT NULL,
    Execution_Price DECIMAL(10,2) NOT NULL,
    Execution_Time TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    Order_ID VARCHAR(50) NOT NULL,
    DematAccId VARCHAR(50) NOT NULL,
    FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID),
    FOREIGN KEY (DematAccId) REFERENCES DematAccount(DematAccId)
);

-- 6. WATCHLIST
CREATE TABLE Watchlist (
    Watchlist_ID VARCHAR(50) PRIMARY KEY,
    Watchlist_Name VARCHAR(100) NOT NULL,
    UserId VARCHAR(50) NOT NULL,
    FOREIGN KEY (UserId) REFERENCES AppUser(UserId)
);

CREATE TABLE Watchlist_Contains (
    Watchlist_ID VARCHAR(50) NOT NULL,
    Ticker_Symbol VARCHAR(20) NOT NULL,
    PRIMARY KEY (Watchlist_ID, Ticker_Symbol),
    FOREIGN KEY (Watchlist_ID) REFERENCES Watchlist(Watchlist_ID),
    FOREIGN KEY (Ticker_Symbol) REFERENCES Financial_Instruments(Ticker_Symbol)
);

-- 7. TRANSACTIONS
CREATE TABLE Financial_Transaction (
    Transaction_ID VARCHAR(50) PRIMARY KEY,
    Transaction_Type VARCHAR(50) NOT NULL,
    Amount DECIMAL(15,2) NOT NULL,
    Payment_Method VARCHAR(50),
    Status VARCHAR(20),
    TimeStamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    TradingAccId VARCHAR(50) NOT NULL,
    BankAccountNo VARCHAR(30),

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FOREIGN KEY (TradingAccId) REFERENCES TradingAccount(TradingAccId),  
FOREIGN KEY (BankAccountNo) REFERENCES LinkedBankAccount(BankAccountNo)  
);
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